Lecture 1

MATH 327

Date: Sep. 2, 2026
Instructor: Manish Patnaik

Contents

1. Axioms for a group 2
2. Associativity, cyclic groups, and order 2
3. Examples 3

1. Axioms for a group

To define a group, we begin with a set G equipped with a *composition law*

$$\cdot\colon G \times G \longrightarrow G, \qquad (g,h) \longmapsto g \cdot h,$$

and a distinguished element $e \in G$, called the *identity*. Thus the basic data are written $(G, \cdot, e)$. We often suppress the dot and write gh in place of $g \cdot h$.

Definition 1 (Group)**.** The data $(G, \cdot, e)$ form a *group* if the following three conditions hold.

(1) Associativity. For all $g, h, k \in G$,

$$g \cdot (h \cdot k) = (g \cdot h) \cdot k.$$

(2) Identity. For every $g \in G$,

$$e \cdot g = g \cdot e = g.$$

(3) Inverses. For every $g \in G$, there exists $h \in G$ such that

$$h \cdot g = e \qquad \text{and} \qquad g \cdot h = e.$$

The first equality says that h is a *left inverse* of g; the second says that it is a *right inverse.* So inverse really means 'double sided' inverse.

Associativity does *not* say that

$$g \cdot h = h \cdot g.$$

An operation satisfying this latter identity for all elements is called *commutative.*

Definition 2 (Abelian group)**.** A group whose operation is commutative is called *abelian.*

It is also useful to observe the following simple result.

Proposition 1 (Uniqueness of inverses)**.** *The element h in axiom **(3)** is unique.*

Proof. Suppose h_1 and h_2 both satisfy axiom **(3)** for g. Then

$$h_1 g = h_2 g = e.$$

Multiplying on the right by h_2 and using associativity gives

$$(h_1 g)h_2 = (h_2 g)h_2, \qquad \text{hence} \qquad h_1(gh_2) = h_2(gh_2).$$

Since $gh_2 = e$, it follows that $h_1 = h_2$. We write the unique inverse of g as g^{-1}. □

2. Associativity, cyclic groups, and order

Associativity allows an unparenthesized product such as

$$g_1 g_2 \cdots g_n$$

to have an unambiguous meaning; it does not allow us to change the order of the factors.

Because of associativity, the expression $g_1 g_2 \cdots g_n$ is unambiguous, regardless of how parentheses are inserted. For $x \in G$ and a positive integer n, we write

$$x^n = \underbrace{x \cdot x \cdots x}_{n \text{ factors}}.$$

If G is finite as a set, its cardinality $|G|$ is also called the *order* of G. If G is infinite, we write $|G| = \infty$ and say that G has infinite order.

3. Examples

Example 1 (The integers under addition)**.** The triple

$$(\mathbb{Z}, +, 0)$$

is a group. Its composition law is

$$\mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}, \qquad (m, n) \longmapsto m + n.$$

The identity element is 0, and the inverse of m is $-m$.

Example 2 (The integers under multiplication: a non-example)**.** The triple

$$(\mathbb{Z}, \times, 1)$$

is *not* a group, although it is a semigroup. Multiplication is associative and 1 is an identity, so axioms **(1)** and **(2)** hold. Axiom **(3)** fails because most integers do not have multiplicative inverses in $\mathbb{Z}$.

Example 3 (Integers modulo n)**.** Let n be a positive integer. The set of residue classes modulo n is

$$\mathbb{Z}/n\mathbb{Z} = \{\overline{0}, \overline{1}, \overline{2}, \ldots, \overline{n-1}\}.$$

For example,

$$\overline{0} = \{0, \pm n, \pm 2n, \pm 3n, \ldots\},$$
$$\overline{1} = \{1, 1 \pm n, 1 \pm 2n, 1 \pm 3n, \ldots\}.$$

Every integer can be written $m = an + r$ with $0 \leq r < n$, and then $\overline{m} = \overline{r}$. Addition is defined by

$$\overline{m} + \overline{\ell} = \overline{m + \ell}.$$

Thus

$$(\mathbb{Z}/n\mathbb{Z}, +, \overline{0})$$

is a group, called the *cyclic group of order n*.

Definition 3 (Cyclic group)**.** A group G is *cyclic* if there is an element $x \in G$ such that

$$G = \{x^k : k \in \mathbb{Z}\}.$$

In this case, x is called a *generator* of G.

In the preceding examples that are groups, the group operation is both associative and commutative.

Example 4 (The general linear group)**.** The set

$$\mathrm{GL}_n(\mathbb{R}) = \{A : A \text{ is an invertible } n \times n \text{ matrix with entries in } \mathbb{R}\}$$

is a group under matrix multiplication. Its identity is the identity matrix

$$I_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}.$$

Thus the group is written $(\mathrm{GL}_n(\mathbb{R}), \cdot, I_n)$, and the composition law sends (A, B) to AB.

Example 5 (The permutation group)**.** The permutation group is

$$S_n = \{\text{all permutations of } \{1, 2, \ldots, n\}\}.$$

Its group operation is composition of permutations.